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End Semester Examination 2025

Name of the Course: B. Tech, EE
 Name of the Paper – Power System-I
 Paper Code: TEE 501

Semester: 5th

Time: 3Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Discuss the advantages of high transmission voltage	CO1
(b)	Compare the volume of conductor material required for a d.c. 3-wire system and 3-phase, 3-wire system on the basis of equal maximum potential difference between one conductor and earth. Make suitable assumptions.	
(c)	A 2-wire feeder carries a constant current of 250 A throughout the year. The portion of capital cost which is proportional to area of X-section is Rs 5 per kg of copper conductor. The interest and depreciation total 10% per annum and the cost of energy is 5P per kWh. Find the most economical area of X-section of the conductor. Given that the density of copper is 8.93 gm/cm^3 and its specific resistance is $1.73 \times 10^{-8} \Omega \text{ m}$.	
Q2		
(a)	Derive the approximate of mechanical sag for unequal supports	CO2
(b)	A transmission line has a span of 200 metres between level supports. The conductor has a cross-sectional area of 1.29 cm^2 , weighs 1170 kg/km and has a breaking stress of 4218 kg/cm^2 . Calculate the sag for a safety factor of 5, allowing a wind pressure of 122 kg per square metre of projected area. What is the vertical sag?	
(c)	The towers of height 30 m and 90 m respectively support a transmission line conductor at water crossing. The horizontal distance between the towers is 500 m. If the tension in the conductor is 1600 kg, find the minimum clearance of the conductor and water and clearance mid-way between the supports. Weight of conductor is 1.5 kg/m . Bases of the towers can be considered to be at water level.	
Q3		CO3
(a)	Derive the relation between string efficiency and voltage distributions across the disc of suspension type insulators. Assume three discs in the systems.	
(b)	Each line of a 3-phase system is suspended by a string of 3 identical insulators of self-capacitance C farad. The shunt capacitance of connecting metal work of each insulator is $0.2C$ to earth and $0.1C$ to line. Calculate the string efficiency of the system if a guard ring increases the capacitance to the line of metal work of the lowest insulator to $0.3C$.	
(c)	A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency.	

Q4		CO4
(a)	Deduce an expression for line to neutral capacitance for a 3-phase overhead transmission line when the conductors are unsymmetrical placed but transposed	
(b)	A 3-phase, 50 Hz, 132 kV overhead line has conductors placed in a horizontal plane 4 m apart. Conductor diameter is 2 cm. If the line length is 100 km, calculate the charging current per phase assuming complete transposition.	
(c)	The three conductors of a 3-phase line are arranged at the corners of a triangle of sides 2 m, 2.5 m and 4.5 m. Calculate the inductance per km of the line when the conductors are regularly transposed. The diameter of each conductor is 1.24 cm.	
Q5		CO5, CO6
(a)	Describe the construction of underground cable	
(b)	What do you mean by grading of cable. Discuss capacitance grading.	
(c)	A single-core cable has a conductor diameter of 1 cm and insulation thickness of 0.4 cm. If the specific resistance of insulation is $5 \times 10^{14} \Omega\text{-cm}$, calculate the insulation resistance for a 2 km length of the cable.	